

An investigation of emerging COVID-19 research trends and future implications for LIS field: A bibliometric mapping and visualization

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Abstract

The COVID-19 epidemic has proved a nightmare for human beings living all across the globe. It has severely affected all segments of human life. Consequently, enormous research has been conducted to assess and counter this global pandemic. This study **reports finding of a bibliometric analysis on COVID-19 related literature published in the Library and Information Science journals, to identify current research trends and to suggest future directions for further research.** It reveals a substantial increase in LIS publications on COVID-19 in the year 2020. The findings of the study are presented in two sections: first, the research productivity analysis provides an overview of the prominent authors, sources, institutions, and countries that published about COVID-19 in the LIS field. The second section offers a co-word thematic analysis of the significant emerging themes that provide an insight into current challenges faced by libraries and suggest future trends for research and practice in post-COVID era. The study exposes 5 major research themes and 11 sub-themes.

Keywords

Bibliometrics, corona-viruses, COVID-19, global pandemic, library and information science research, research productivity

Introduction

The outbreak of coronavirus or COVID-19 has toppled the daily lives of human beings living all over the world (Verma and Gustafsson, 2020). It has severely affected on the health of all age groups, including infants, young, and the old (Donthu and Gustafsson, 2020). Because of the massive spread of this virus from one human to other on a large scale, in March 2020, the WHO declared it a global

pandemic. As of February 2021, about 110,384,747 people have been infected globally and 2,446,008 have lost their lives (World Health Organization, 2021). Policies such as

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complete or smart lockdown of cities and “social distancing” or “stay-at-home” have been implemented in almost all countries (Leite et al., 2020). Despite of the precautionary measure taken all across the world, the danger still exists, and the world is under the threat of the second wave of this pandemic (TRT World, 2020). COVID-19 has not only caused an extensive damage to the public health, but it has also severely damaged the business, education, and economy of the world (Rodrigues et al., 2020; Yang et al., 2020). This dreadful disease has gained significant attention of the researchers and various studies have been conducted to look into the different aspects of COVID-19. Research work is important in this hard time, as it helps how to tackle this novel epidemic (Raju and Patil, 2020). Consequently, research organizations and academic institutions have positively responded to this crisis by freely sharing the findings and research data of studies relevant to this pandemic. Many journals have also been offering free access to their research related to COVID-19 to aid the diagnosis, treatment, and precautionary measures against the virus (Gong et al., 2020). Bibliometric studies help researchers in measuring the research output in a specific field. It also helps in finding the evolving areas of research; measures the impact of research through citations analysis, carries the research ahead, and help policy makers on focusing and funding a specific area of research (Raju and Patil, 2020). Many bibliometric studies including Aristovnik et al. (2020), Chahrour et al. (2020), Darsono et al. (2020); El Mohadab et al. (2020), Fan et al. (2020), Hossain (2020), Huded and Balutagi (2020), Nasab (2020), Oh and Kim (2020), Pernitez-Agan et al. (2020), Rodrigues et al. (2020), Zyoud and Al-Jabi (2020), Shamsi et al. (2020), and Tao et al. (2020) have been conducted on COVID-19 publication patterns however; no study has been conducted so far to look into the contribution of Library and Information Science (LIS) journals toward publishing on COVID-19. This identifies a major gap in the literature that what exactly are LIS journals positioned to tell us about COVID-19? The current study is an effort to fill this gap.

Literature review

Bibliometric studies have been widely conducted in current years, in order to analyze the scientific contributions made in various areas of research (Patiño-Barbosa et al., 2016). The review of the current literature reveals that several researchers have conducted a good range of bibliometric studies covering various aspects of the COVID-19. Some key findings of these studies are discussed in this section. The study of Gong et al. (2020) is one of the premier bibliometric studies on COVID-19. They analyzed over 1500 papers published from January 1, 2020 to March 8, 2020 on the topic of COVID-19 which were indexed in PubMed and China National Knowledge Infrastructure databases. Their findings revealed a high growth trend of

publications on COVID-19. Moreover, most of the publications were from China and an extraordinary cooperation and information sharing was also reported among the scientific community at global level. Verma and Gustafsson (2020) conducted a subject specific study to bibliometrically analyze the literature published in areas of business and management. Findings of their study revealed that within four and half months of the covid-19 pandemic 107 documents had been published in Scopus and WoS indexed databases. They also identified the most productive researchers and areas and sub-areas of COVID-19 research in business and management fields and concluded that the crisis of COVID-19 will have long lasting effects on all businesses of the world.

Raju and Patil (2020) investigated about the contribution of Indian publications on COVID-19 disease in WHO database. Their study revealed a substantial contribution of Indian publications in WHO database. Authors of the AIIMS institute were identified as the most prolific contributors whereas the AIIMS institute identified as the most productive institution while the Indian journal of medical research was the most productive journal. Yang et al. (2020) analyzed the contribution of Chinese researcher’s research publications on preparing the medicine for COVID-19 through a bibliometric analysis. They identified a good number of articles 309 had been published on the subject, including 61 journals, 1441 authors, and 277 institutions. The study of Hossain (2020) evaluated the 371 articles published during 2019–2020 on COVID-19 topic which were indexed in WoS database. He determined from the results of his study that COVID-19 research has shown good progress in different areas, however, he emphasized on the need for more research on socioeconomic determinants of health, and need for collaboration in research among global nations to mobilize collective resources in understanding COVID-19. Chahrour et al. (2020) explored the PubMed and WHO databases and analyzed the 564 publications on COVID-19 published from December 2019 to March 2020. He shortlisted and analyzed only those studies which were observational and interventional in nature. The findings of his study revealed China as the largest contributor in terms of number of publications.

Hamidah et al. (2020) bibliometrically analyzed the 3513 Scopus indexed articles published on COVID-19. Their findings identified a significant increase of publications on COVID-19 during 2019–2020. China was on the top with most publications while the keywords of coronavirus and pandemic were most widely discussed by the researchers. Lou et al. (2020) searched in the PubMed, a database of medical publications by using the keyword “COVID-19” and analyzed 183 publications published from January 2020 to February 2020. Findings of their study exposed that most of the authors of the publications were affiliated from hospitals while, “journal of medical virology” published the most of publications on COVID-19. Yu et al. (2020) reviewed the 2019–2020 published literature on the topic of COVID-19

by searching in the Web of Science core collection database. Their search in WoS database retrieved a total of 3626 records on the topic. They found that most of the studies were descriptive in nature and were focused on three areas, including disease management, clinical features and pathogenesis. British medical journal identified as the highest ranked journal with most publications on COVID-19. In context of the most citations, Christian Drosten from Germany and Chinese Academy of Sciences from China were on the top. Tao et al. (2020) reviewed the publications of the last 20 years published on COVID-19 by searching in the Web of Science database. They retrieved a total of 9760 records on the items published on corona viruses. Analysis of the data identified Journal of Virology as the most reputed title on the topic. The United States of America followed by China were on the top positions with most of the publications. In context of the academic institutions, the University of Hong Kong was on the leading position while, Prof. Yuen and Peiris had made some great achievements in coronavirus research.

The study of Kambhampati et al. (2020) reported an exceptional growth of publications on COVID-19 since the breakout of the disease in November 2019. They also reported that the ratio of publications on COVID-19 is much more as compared with any other disease. Fan et al. (2020) conducted a bibliometric study in the early days of the COVID-19 pandemic to explore the variation between English and Chinese language medical journals. Findings of their study revealed that a majority of the researchers and their affiliated institutions of these publications were from Chinese region. Moreover, the keywords of the English language publications were more wide-ranging than those of the Chinese publications. Both, the English and Chinese publications worked as useful sources to disseminate valuable information regarding COVID-19. However, a difference of the audience was observed in both publications as in English publications the audience was medical practitioners and scientists whereas in Chinese publications the local level medical practitioners were given essential information on how to tackle the COVID-19. Moreover, their results also showed that COVID-19 disease and the policies of social distancing and lockdowns had resulted great effect on the mental health of the people. Consequently, during this period many people have become patients of anxiety and depression.

Zambrano et al. (2020) bibliometrically reviewed the 223 psychology related publications on COVID-19 by using the data available in Scopus database. They wanted to examine the effects of isolation and social distancing on the mental health of general public. Their findings revealed that a majority of the studies on the topic were conducted in China. The researchers of these studies had suggested various strategies how to persuade individuals to comply with social distancing. Aristovnik et al. (2020) conducted an extensive bibliometric study on COVID-19 related publications indexed in the Scopus database. Their search retrieved a total of 16,866 records. Their findings disclosed a lack of

publications on COVID-19 in the field of social sciences. Hossain et al. (2020) bibliometrically measured the scholarly literature published in the domain of application of artificial intelligence technologies in COVID-19 research. They analyzed 105 articles retrieved from the MEDLINE database. Their findings showed that a majority of the articles were based on collaborative work and most of the contributors were from USA and China, while developing countries had very less contribution in this regard.

Bibliometric studies have also been conducted in the field of Library and Information Science by a number of researchers. Nath and Jana (2020) recently conducted a statistical analysis of a journal *Annals of Library and Information Studies*. Similarly, Kannan and Thanuskodi (2019) conducted a bibliometric study of another LIS journal *Library Philosophy and Practice*. Okeji (2019) also carried out a bibliometric analysis of the research publications authored by the librarians in Nigeria. His findings revealed that a few librarians are engaged in research publishing in Nigeria. However, the current review of the literature shows that there is a lack of work on subject specific bibliometric studies on COVID-19 published in LIS journals. Therefore, this study particularly focuses on the literature published on COVID-19 in Library and Information Science journals.

Scope of the study

The novel coronavirus (2019-nCoV) or the severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) spread faster than its two ancestors did, the SARS-CoV in 2002 and Middle East respiratory syndrome coronavirus (MERS-CoV) in 2012 (Singhal, 2020). This study covers two main findings: the influential LIS-COVID research, the LIS research focus related to COVID. In the first section, we discuss the prominent aspects of the LIS literature published on SARS, MERS-CoV, and COVID. SARS and MERS-CoV gained worldwide attention but were less fatal and remained under regional control compare to COVID-19. Therefore, the effects on the library field remained almost unnoticed. However, the COVID-19, being a pandemic and its global impact on all aspects of human life and future, put libraries in a critical situation. Like other disciplines, LIS researchers and practitioners quickly responded to this situation and devised plans to continue services during the pandemic's socioeconomic effects. It is difficult to estimate the damage and predict a post-COVID strategy for the LIS field as the pandemic is still in progress. However, it is evident that library activities' shape and scope will not be the same as pre-COVID times. The society had evolved rapidly during 2020; thus, the needs and expectations have changed too. In the second section, we used text-mining techniques to visualize the focused areas in LIS-COVID research. It is to highlight the main issues posed by the pandemic and how libraries have responded to it. Finally, we discussed the post-COVID implications for the LIS field and indicated future research

directions. We hoped that this study would be a starting point of discussion for the library professionals and academicians to estimate the post-COVID era's affordances.

Research questions

This study explores the COVID-19 related publications published in LIS journals to identify:

1. What are the research trends in LIS journals related to COVID-19?
2. Which organizations are most productive, in terms of COVID-19 research?
3. What are the authorship patterns in COVID-19 research?
4. Which LIS journals are most productive in COVID-19 research?
5. Which are the most highly cited publications in COVID-19 research?
6. Which are the most prolific authors in COVID-19 research?
7. Which are the most productive countries in terms of COVID-19 publications?
8. What are the emerging research themes on COVID-19 research in LIS journals?

Research methods and procedures

The researchers conducted this study by applying bibliometrics methods to quantitatively analyze the data and to highlight research productivity and co-word analysis of LIS-COVID research. For this purpose, one of the reliable and large bibliographic data source **Scopus was used to retrieve data on the topic**. Scopus does not provide any subject category for Library and Information Science (LIS) discipline. On the contrary, Scopus allocate subject code to journals according to All Science Journal Classification (ASJC) and assigns 3309 code to Library and Information Science journals. Therefore, the researchers used 3309 code to retrieve 295 journals related to Library and Information Science discipline. The ISSN of all print and electronic journals were identified, and a query was formulated to retrieve documents published in the journals by joining all ISSN with OR operator then AND operator for all possible keywords for coronavirus or COVID-19. Therefore, the following query was executed in the search window of the Scopus database which retrieved 501 records on January 21, 2021. TITLE-ABS-KEY (coronavirus* OR covid19 OR "covid 19" OR covid-19 OR ncov-* OR hcov-* OR sars-cov* OR "severe acute respiratory syndrome" OR mers-cov* OR "Middle East Respiratory Syndrome" OR "corona virus"). All the records were imported into the EndNote software, and a duplication check was applied on the author, title, and year match, which resulted in no duplicate records. Hence, 501 documents were finalized to use in the study. The

researchers used MS Access, MS Excel, Notepad++, Biblioshiny, HistCite, and VOSviewer software's for data analysis and visualization purposes.

The researchers also conducted a co-word analysis by using VOSviewer (See Figure 1) to identify the emerging research trends of the COVID related research in the LIS field. Co-word analysis is one of the methods that use to identify themes, sub-themes, and their relationships. Co-word analysis is a text-mining technique where co-occurrences of the key-words in publications is calculated and later clusters are created based on themes and sub-themes (Ravikumar et al., 2015; van Eck and Waltman, 2010). This analysis helps to identify the main topics and emerging trends in the given field and how themes relate with other fields (Waltman et al., 2010). The researchers performed the keyword co-occurrences analysis with the VOSviewer software and selected the three—check word range. There were 16 clusters and 191 sub-categories in auto-analysis run by VOSviewer. Team of our researchers discussed and performed manual data cleaning and duplication removal. We also standardized the words (i.e. singular/plural) and spelling (i.e. American/British). After cleaning data, we analyzed the clusters and skim read full text when got confused. We used a beta version of VOSviewer. At the end, we selected 5 main themes and regrouped keywords under 11 sub-themes according to keywords relevance and logical relationships (see Table 1).

Findings and discussion

Research productivity analysis in LIS-COVID research

The year-wise publishing trends related to LIS-COVID research is shown in Table 2. The data portray a lack of interest of LIS researchers on the topic starting from the year 2002 with only 21 publications up to the year 2019. However, the topic got attention of the LIS researchers in the year 2020 with an unprecedented number of publications. The data also highlight a good number of publications in the year 2021 so far by the time of data retrieval for the study. The further analysis regarding total citations received in a year also ranked 2020 as the top year with over 1000 citations, followed by the years 2005, 2004, and 2011.

Most productive organizations in LIS-COVID research

The top 10 most productive organizations have been presented in Table 3. The data was sorted on total publications then on total citations received by an organization. The Universidad de Granada, The University of Sydney, and Universidad de Murcia have equally produced seven publications. But the Universidad de Granada obtained more citations than the other two organizations hence ranked at the top position. Similarly, the National University of Ireland

Table 2. Year-wise publishing trends in LIS-COVID research.

Year	2021	2020	2019	2018	2015	2014	2012	2011	2005	2004	2003	2002
TP	21	459	3	2	1	1	1	4	2	4	2	1
TC	1	1088	8	11	2	22	3	58	87	64	4	4

TP: total publications; TC: total citation.

Table 3. Most productive journals and organizations in LIS-COVID research.

Most productive organizations			Most productive journals in LIS-COVID research		
Organization	TP	TC	Journal Name	TP	TC
Universidad de Granada	7	35	Profesional de la informacion	37	195
The University of Sydney	7	23	Library philosophy and practice	35	5
Universidad de Murcia	7	2	International journal of information management	27	254
National University of Ireland Galway	6	41	Scientific data	25	135
Universitat Pompeu Fabra Barcelona	6	28	Information and learning science	25	19
Delft University of Technology	6	11	Journal of chemical information and modeling	24	270
Hong Kong Baptist University	5	28	Scientometrics	20	105
Universite Paris-Saclay	5	18	Knowledge cultures	20	3
Chinese Academy of Sciences	5	9	European journal of information systems	19	70
Universitat de Barcelona	5	7	Information systems management	18	7

Galway, Universitat Pompeu Fabra Barcelona, and Delft University of Technology also produced six publications, each with a difference in the number of citations received. Interestingly, the last four organizations also equally produced five publications, each with a variation of citations received.

Most productive journals in LIS-COVID research

The list of top10 journals producing research on the topic is shown in Table 3. The analysis ranked the journal “Profesional de la informacion” at the top position with 37 publications followed by “Library philosophy and practice” and “International journal of information management.” The citation analysis of data ranked “Journal of chemical information and modelling” at the top position with the highest 270 citation score followed by “International journal of information management.” The journal “Profesional de la informacion” that secured top position in terms of the number of publications could not maintain its position and ranked at the third position in obtaining the number of citations. There is another interesting finding regarding the journal “Library philosophy and practice,” which was the second top journal in producing publications on the topic but remained at the second last position in making an impact with only a total of five citations.

Authorship pattern in LIS-COVID research

The authorship pattern in LIS-COVID research is presented in Figure 2. The data revealed the authorship pattern ranged from a single to 27 authors. The single-author

pattern remained the most favorite pattern by the LIS researchers, with 179 publications followed by two-author and three-author patterns. A total of 19 publications were produced by authors ranging from 11 to 27 author patterns. The analysis regarding citations highlights very interesting findings. The single-author pattern was also at the top position in terms of obtaining the number of citations followed by a three-author pattern. Simultaneously, the authorship pattern ranging from 11 to 27 authors ranked at the third position in obtaining the total number of citations. The two-author pattern remained at the fourth position in receiving citations on the respective publications.

Document types

The document type in LIS-COVID research is presented in Table 4. The data ranked the document type “Article” as the most preferred format by LIS researchers for sharing their research with colleagues on the topic, followed by “Note” and “Editorial.” The further analysis regarding the citations obtained by each document type also ranked “Article” with more than 1100 citations followed by “Data paper” and “Note.”

Top highly cited articles

The top 10 highly cited articles on LIS-COVID research have been shown in Table 5. The data ranked “Fast identification of possible drug treatment of coronavirus disease-19 (covid-19) through computational drug repurposing study” at the top position with the highest total citations obtained even it was published in the year 2020, followed

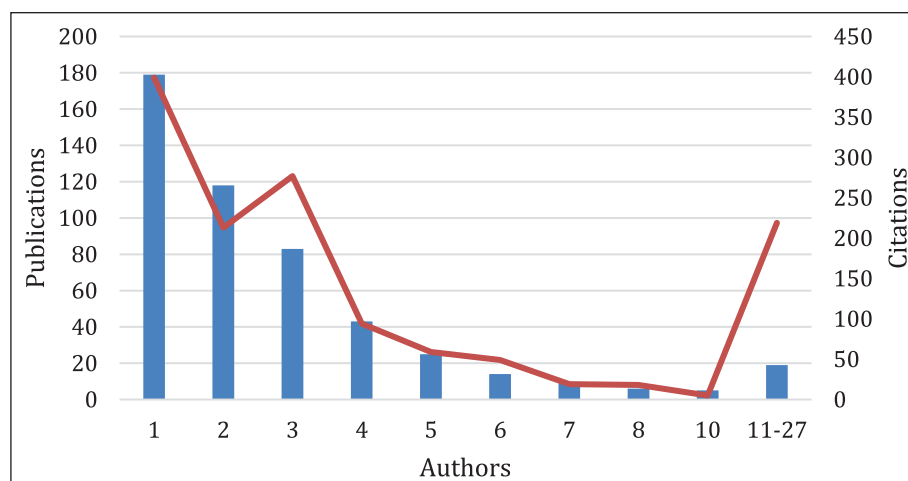


Figure 2. Authorship pattern.

Table 4. Document types.

Document type	Article	Note	Editorial	Review	Data paper	Letter	Erratum
TP	416	26	18	17	16	7	1
TC	1114	70	25	27	96	20	0

by “Virtual screening of novel noncovalent inhibitors for sars-cov 3c-like proteinase” with second-highest citations published in 2005. The further analysis revealed that among the top 10 articles, 3 were published in “Journal of chemical information and modeling.”

Most productive authors

Table 6 presents the most productive authors on LIS-COVID research. The three most prolific authors on the topic belong to Pakistan, with the top position equally maintained by two Pakistani LIS researchers “Bhatti, R.” and “Naeem, S. B.” The further analysis ranked “Conboy, K.” at the top position in terms of obtaining citations followed by “Ali, M.Y.” and “Thelwall, M.”

Author keyword analysis

Figure 3 represents the author keyword analysis with a minimum number of occurrences of a keyword as five. Of the 1499 keywords, 56 met the threshold. The most prominent keywords in the map are COVID-19, coronavirus, pandemic, social media, online learning, fake news, academic libraries, higher education, information literacy, and knowledge management.

Country co-authorship

In this study, the researchers used VOSviewer to visualize international collaboration in the LIS-COVID research.

The map provides valuable information and guides potential researchers in identifying the international collaborators on the topic. Figure 4 presents the collaborative work of countries with minimum number of documents of a country as one, with a minimum number of citations as zero. Out of 95 countries, all met the threshold and were distributed into 11 clusters, and countries with the same color maintain 1 cluster. Each circle represents a country, and the size depicts the number of publications contributed by the country. The larger the size of the circle, the larger will be the contribution of the countries in terms of the publications, and the thickness of the lines between the countries portrays the strength of the collaboration. The most prominent countries on the map include the United States, Spain, France, Germany, Netherland, India, China, Australia, and the United Kingdom.

Country citations

The citation analysis can be taken as when the works of two different authors are simultaneously cited by a third author and then the two authors can be regarded as having a co-citation relationship. Figure 5 highlights the country citations map with the minimum number of documents as one and the minimum number of citations as zero. The map divides the countries into six clusters. The size of the circle indicates the number of publications co-cited, which means the larger the circle, the more publications co-cited. The length of the line between two countries reflects the relationship, which means the shorter the line between two countries shows the closer relationship.

Table 5. Top 10 highly cited articles.

Author	Title	Source	PY	TC
WANG J	Fast identification of possible drug treatment of coronavirus disease-19 (covid-19) through computational drug repurposing study	Journal of chemical information and modeling	2020	88
LIU Z;HUANG C;FAN K;WEI P;CHEN H;LIU S;PEI J;SHI L;LI B;YANG K;LIU Y;LAI L	Virtual screening of novel noncovalent inhibitors for sars-cov 3c-like proteinase	Journal of chemical information and modeling	2005	65
CHIU WT;HUANG JS;HO YS	Bibliometric analysis of severe acute respiratory syndrome-related research in the beginning stage	Scientometrics	2004	60
XU B;GUTIERREZ B;MEKARU S;SEWALK K;GOODWIN L;LOSKILL A;COHN EL;HSWEN Y;HILL SC;COBO MM;ZAREBSKI AE;LI S;WU CH;HULLAND E;MORGAN JD;WANG L;OBRIEN K;SCARPINO SV;BROWNSTEIN JS;PYBUS OG;PIGOTT DM;KRAEMER MUG CASERO-RIPOLLS A	Epidemiological data from the COVID-19 outbreak, real-time case information	Scientific data	2020	59
	Impact of COVID-19 on the media system. Communicative and democratic consequences of news consumption during the outbreak	Profesional de la informacion	2020	38
LAATO S;ISLAM AKMN;ISLAM MN;WHELAN E	What drives unverified information sharing and cyberchondria during the COVID-19 pandemic?	European journal of information systems	2020	27
AEVERMANN BD;PICKETT BE;KUMAR S;KLEM EB;AGNIHOTHARAM S;ASKOVICH PS;BANKHEAD III;BOLLES M;CARTER V;CHANG J;CLAUSS TRW;DASH P;DIERCKS AH;EISFELD AJ;ELLIS A;FAN S;FERRIS MT;GRALINSKI LE;GREEN RR;GRITSENKO MA;HATTA M;HEEGEL RA;JACOBS JM;JENG S;JOSSET L;KAISER	A comprehensive collection of systems biology data characterizing the host response to viral infection	Scientific data	2014	22
HARRIS R;NADINE WATHEN C;CHAN D	Public library responses to a consumer health inquiry in a public health crisis	Reference and user services quarterly	2005	22
MUKHERJEE P;SHAH F;DESAI P;AVERY M	Inhibitors of sars-3clpro: virtual screening, biological evaluation, and molecular dynamics simulation studies	Journal of chemical information and modeling	2011	21
KUMMITHA RKR	Smart technologies for fighting pandemics: the techno- and human- driven approaches in controlling the virus transmission	Government information quarterly	2020	21

Country coupling

Bibliographic coupling is opposite to the co-citation analysis. If two publications cite any third publication, then the two publications will be considered bibliographically coupled. Figure 6 presents the country coupling map with the minimum number of documents as one, the minimum number of citations as zero, and all met the threshold. The map divides the countries into four clusters. The size of the circle depicts the number of publications coupled, which

means the larger the circle, the more the coupling of documents found. The length of the line between countries reflects the relationship, which means the short the line between two countries indicates the close relationship.

Thematic/co-word analysis

Library sector is greatly affected by COVID, as a yearlong lockdowns and social distancing restrictions put the libraries in an unexpected situation to continuation of operations

Table 6. Most productive authors.

Author Name	Affiliation	Country	TP	TC
Bhatti, R.	The Islamia University Bahawalpur	Pakistan	4	17
Naeem, S.B.	The Islamia University Bahawalpur	Pakistan	4	17
Conboy, K.	Lero, National University of Ireland Galway, Galway	Ireland	3	27
Thelwall, M.	University of Wolverhampton	UK	3	19
Ratzan, S.C.	City University of New York	USA	3	6
Ali, M.Y.	The Aga Khan University, Karachi	Pakistan	2	20
Auffray, C.	University of Luxembourg	Luxembourg	2	12
Balling, R.	University of Luxembourg	Luxembourg	2	12
Aguilar, S.J.	University of Southern California, Los Angeles, CA	USA	2	2
Adams, A.L.	University Libraries, California State University, Hayward, CA	USA	2	

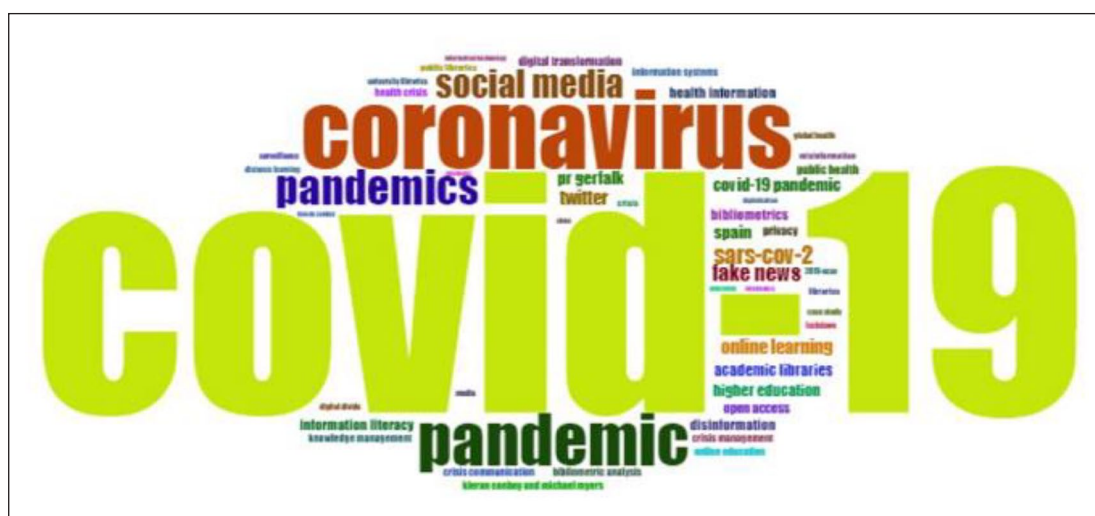


Figure 3. Author keyword analysis.

and services. In this section, we present the co-word occurrence analysis under 5 themes and 11 sub-themes.

Cluster 1: Libraries and services

Our first cluster is about the libraries and services. Libraries as public service institutions were enlisted among places to remain close for most of year 2020 for physical services. Different types of libraries (e.g. university libraries, public libraries, elementary school, K-12,) appeared during the co-occurrence analysis. Fortunately, many libraries (e.g. university libraries and research libraries) were efficient to respond by extending their online services. However, school libraries and public libraries struggle to initiate or to continue online service due to user connectivity issues. The shift of services during lockdown and physical distancing was in the areas of information services, digital services, reference services, public relations, interlibrary loan, and library cooperation.

During lockdown, many professionals virtually connect with peers from around the globe forming communities of practices. Library associations offered various free

online workshops, seminars, discussions, and meetups to offer support and experience sharing to cope with pandemic crisis. A feature of these virtual meetings was open participation from the librarians of developing world and possibilities of connecting and learning from practices of developed world librarians. Through formation of this loose form of community of practice (CoP), librarians were able to stay motivated and productive during times of uncertainty.

Cluster 2: Library management

Due to unexpected and uncertain events caused by COVID-19 spread, library management faced various challenges. The effects of pandemic are still in progress and will cause further challenges. Although, it is possible to estimate the future directions and to predict course of action of libraries, professional and library schools. Libraries can play multiple roles in emergency or disaster situations, including as “institutional supporters, collection managers, information disseminators, internal planners, community supporters, government partners,

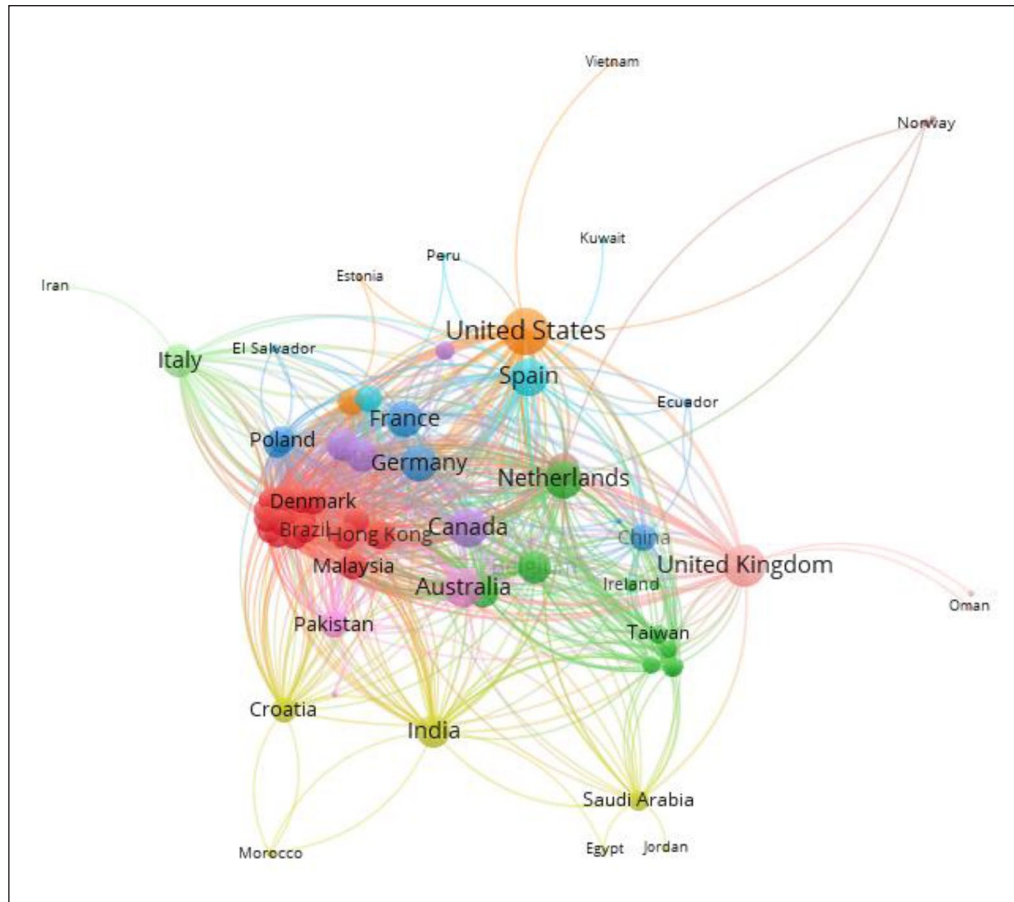


Figure 4. Country co-authorship.

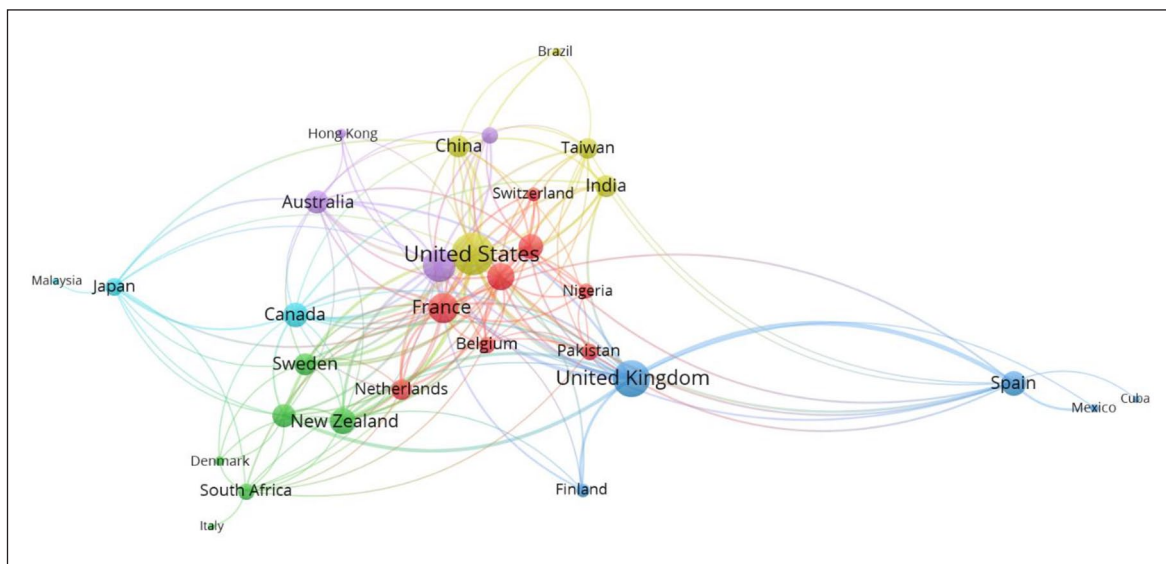


Figure 5. Country citations.

educators and trainers and information community builders” (Featherstone et al., 2008; Koscijew, 2021). Keyword analysis showed various interesting aspects related to library management. It presented COVID related

challenges, for example, lockdown, impact, distribution, crisis, business continuity and sustainability. Pandemic significantly disrupted library services and caused a state of emergency for librarians. Libraries have responded to

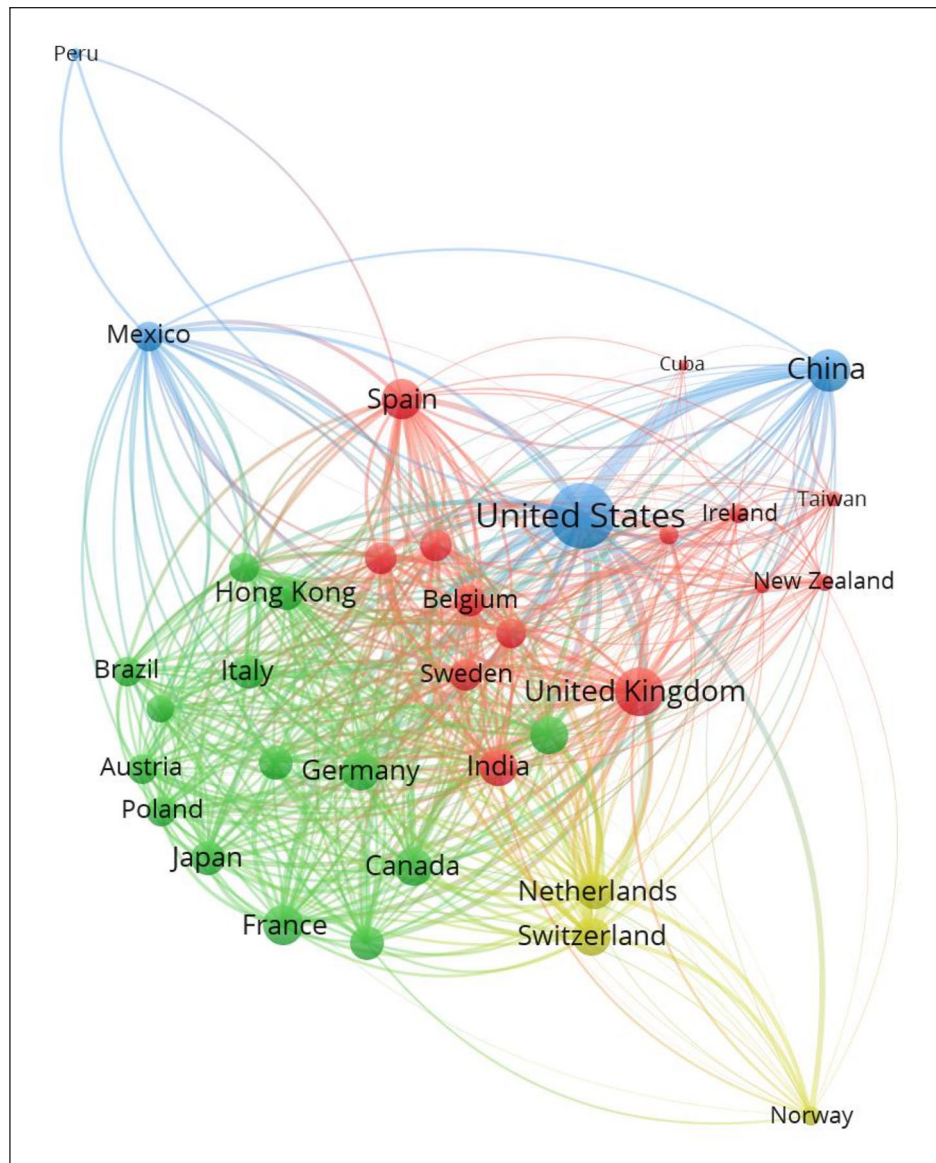


Figure 6. Country coupling.

this extraordinary situation by offering their services online through digital access yet remaining closed to the public (Samanta, 2020). Libraries worldwide are facing hard choices around which services to offer and how, ranging from minimal restrictions to full closure, reopening and which services to continue in times of uncertainty. However, libraries around the world are trying to offer any service (electronic if not physical), in pursuit of staying normal and continuing as usual (IFLA, 2020). The Australian Library and Information Association (ALIA) stated, “libraries have been highly agile, adapting and embracing new technologies, delivering online forums and making electronic resources widely available. As a sector, libraries have proved to be proactive and exceptionally resilient throughout the pandemic” (Cleave and Geijsman, 2020). Pandemic has awoken human society’s natural

resilience, and social evolution is as rapid as the virus’s speed. Similarly, libraries were quick to make sense of the situation, showing agility and enhancing their affordances by experimenting with remote work and collaboration. As libraries re-opened, many offered real-time reference services, as well as seamless and contactless curbside service, providing a sense of control and continuity amongst the chaos (Goddard, 2020).

Cluster 3: Libraries and education

Learning helped people around the globe to stay positive and normal during the lockdowns of 2020. The learning process during 2020 not only grew but showed digital transformation in developed and developing world. Educational institutes from tertiary education to kindergarten went

virtual for enrolled students. On the other hand, adults and teens turned to self-learning from online courses, skill development, and virtual meeting with communities of practice. Online teaching and learning in a way that ensures accessibility and equity for all, issues connecting to online pedagogy and how to prepare students with the information and digital literacy competences required for the new online learning “normal” (Martzoukou, 2021). Libraries as the heartbeat of educational institutions have been hard hit (Oyelude, 2020). Online education and online learning related terminologies were mostly found among LID-COVID literature. We categorize these keywords into two thematic categories - online learning and information literacy. Different types of libraries responded differently to online learning requirements. Academic libraries offered support of online information literacy instructions to support the online learning. The public libraries also demonstrated value to their communities. They have enabled their patrons to not only resume their lives, but to help them learn and grow (Goddard, 2020).

Cluster 4: Libraries and technology

Fortunately, most libraries worldwide have some level of access to e-resources or platforms to offer services. Other libraries rushed to adopt the online platform to provide virtual support to their users. The themes emerged from the co-word analysis showed that digital collections, digital technologies, repositories, and digital transformation. Kids being home from school and learning virtually, it is no surprise that ebook readership skyrocketed amongst YA and Juvenile readers with an 87% increase from last year (Goddard, 2020).

Cluster 5: Libraries and information society

In the last cluster, we grouped themes related to the role of libraries in the information society. The subthemes and related keywords cover—digit divide, communication, misinformation, digital rights, and responsibilities. Libraries work to support these aspects of digital society as services providers and as advocates of user awareness.

Digital divide. One of the aspects of the information society is access to digital information. Libraries are an advocate of equal access opportunities to everyone. During the pandemic, libraries worked to renew their mission to help learners in the online space become information-rich and digitally competent and empower them to overcome digital divides and inequalities (Martzoukou, 2021). Many libraries, institutional repositories and publishers made agreements to open up material to the general public to bridge the digital divide during pandemic (Tsekea and Chigwada, 2021). There are various concerns for sustaining digital access for users post-COVID, as libraries might

face financial cuts that would affect the cost of digital resources infrastructure (Mbambo-Thata, 2021).

Communication. COVID-19 has also changed the boundaries between formal and informal communication, challenging existing codes of practice and conduct in online working settings (Martzoukou, 2021). The co-word analysis of LIS-COVID research showed an interesting range of communications. LIS researchers discussed digital and social media related keywords, for example, digital journalism, news consumption, health communication, risk communication, scholarly communication, scientific communication, corporate communication, crisis communication, institutional communication, and political communication. Communication served extreme purposes during the pandemic that ranged from slowing down the virus’s spread to avoiding social isolation.

Misinformation. Misinformation was among the most significant threat during the pandemic that created socio-economic, ethnic, and political challenges to human civilization. Especially with unprecedented misinformation, rumors, conspiracy theories, and fake news being spread daily, the librarian as information broker needs to remain relevant in the virtual learning environments (Ajibade and Mutula, 2021; Ladan et al., 2020; Tsekea and Chigwada, 2021). Libraries association also guided librarians about the importance of action against misinformation. ALA stated that as stewards of accurate information, we connect library users with local public health resources and services. It ensures that library users and non-users know about the valuable resources libraries have to offer and curated lists to COVID-19 news and examples of best-practice responses to cope with the misinformation during pandemic (ALA, 2020; Koscieljew, 2021). Our analysis showed some other themes related to information disorder, for example, infodemic, post-truth, credibility, trust, disinformation, fact checking from the LIS-COVID literature. Information disorder paradigm is a comprehensive model for understanding the scope and depth of misinformation, disinformation and mal-information for libraries (Qutab et al., 2019).

Digital rights and responsibilities. Libraries play their role in the information society as an access point to information, through advocacy on digital rights and responsibilities and offering teaching and guidance. During the pandemic, tracking apps and maintaining a log of mobility became essential to prevent or reduce the virus’s spread. Apart from its benefits, various concerns over data collection, data protection, surveillance, digital ethics, privacy, digital ethics, fairness, and human rights raised by the public and documented by researchers.

Conclusions and recommendations

Research plays a pivotal role in order to understand a problem and to design strategies to counter it. The COVID-19 pandemic also demands grave attention of the scientists and researchers involved in the scholarly communication. This study concludes that a good range of research on COVID-19 has been published in LIS journals. Universidad de Granada, of Spain identified as the most productive institution in term of publications and citation counts. The tendency of single-authorship pattern remained dominant while, the document type “Article” remained the most preferred format among LIS researchers for sharing of research. “Profesional de la Informacion” identified as the most productive journal, while an article “Fast identification of possible drug treatment of coronavirus disease-19 (COVID-19) through computational drug repurposing study” received the highest number of citations. Two authors from Pakistan identified as the most prolific authors on COVID-19 research.

The co-word analysis of the study showed a range of concerns and challenges that libraries faced during the pandemic. Some of these prominent challenges include nationwide lockdowns, social distancing restrictions and spread of misinformation among the masses. The physical library services remained severely interrupted due to these limitations, however, libraries managed to offer online services to their users. Misinformation was among some of the other challenges that librarians faced the most during the pandemic. Therefore, librarians need to keep themselves updated about the authentic information sources to counter any of the ongoing spread of rumors and disinformation. On the other end, these challenges have created an opportunity for libraries to evolve and prepare to respond to crisis to play their role in the post-COVID era. Libraries have been forced to see new options, explore new technologies, and evolve with digital society. There can be structural changes that will impact the library personnel, infrastructure, resources, and services. Overall librarians and libraries have positively responded to the COVID-19 pandemic by aligning their roles, and services with required circumstances. Nonetheless, need for more collaborative research and consolidated strategies can further aid libraries to better serve the library users in odd times. The researchers propose further research in the following sub-themes: including crisis management in libraries, work from home, sustainable development goals, sustainable strategy, infodemic, misinformation, disinformation, fact-checking, fake news, e-learning, online education. The study also emphasizes the need that the global knowledge base should be strengthened for evidence-based decision-making preventing and addressing the COVID-19 outbreak around the world.

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